JSES International 9 (2025) 1781–1792

Contents lists available at ScienceDirect

JSES International

journal homepage: www.jsesinternational.org

Trends of shoulder arthroplasty and rotator cuff repair in Japan: national database analysis

Masataka Minami, MD, PhD[a,b,*], Masamitsu Kido, MD, PhD[b], Yoshikazu Kida, MD, PhD[b], Ausberto R. Velasquez Garcia, MD[a,c], Kenji Takahashi, MD, PhD[b], Shawn W. O'Driscoll, MD, PhD[a]

[a]Department of Orthopedic Surgery, Mayo Clinic, Rochester, MN, USA
[b]Department of Orthopaedics, Graduate School of Medical Science, Kyoto Prefectural University of Medicine, Kyoto, Japan
[c]Department of Orthopedic Surgery, Clinica Universidad de los Andes, Santiago, Chile

ARTICLE INFO

Keywords:
Rotator cuff tear
Rotator cuff repair
Shoulder arthroplasty
Reverse total shoulder arthroplasty
National database
Epidemiology

Level of evidence: Level IV; Descriptive Epidemiology Study Using Large Database

Background: Reverse total shoulder arthroplasty was introduced to Japan in 2014, principally to treat cuff tear arthropathy and irreparable massive rotator cuff tears. Its indications have expanded significantly worldwide since then. This study aimed to analyze the annual trends in shoulder arthroplasties and rotator cuff repairs (RCRs) from 2014 to 2022 and to examine differences by sex and age group using a publicly available national database.
Methods: We conducted a retrospective analysis using data from the National Database of Health Insurance Claims and Specific Health Checkups provided by the Ministry of Health, Labour, and Welfare in Japan. The surgical categories analyzed included shoulder arthroplasty and RCR between 2014 and 2022. The annual number of surgeries was reported according to sex and age group. Age-specific incidence was calculated per 100,000 people using age group population data. Descriptive statistics and the independent *t*-test were used for comparisons by sex and age, and linear regression analysis was applied to assess trends.
Results: Shoulder arthroplasties increased dramatically from 1,246 in 2014 to 5,439 in 2022. From 2014 to 2022, the number of RCRs did not decrease; in fact, they continued to increase steadily from 17,553 to 21,183, except for a dip in 2020. Shoulder arthroplasties were nearly twice as common in females, whereas RCRs were more frequent in males. Shoulder arthroplasties peaked in the 75-79 age group, with an average age of 78 years, whereas RCRs were most frequent in the 70-74 age group, averaging 67 years.
Discussion and Conclusion: Despite the substantial increase in shoulder arthroplasties, the number of RCRs has also slightly increased. These findings demonstrate that reverse total shoulder arthroplasty did not simply replace RCRs but complement conventional RCR in the treatment spectrum of shoulder disorders. The findings indicated that females had a higher prevalence of shoulder arthroplasty procedures, whereas males were more likely to undergo RCR. In addition, the patients who underwent shoulder arthroplasty tended to be older than those who underwent RCR. These observations may be attributed to sex-specific differences in the progression of shoulder conditions and the accessibility of various treatment approaches or possibly a difference in indications for surgery in females and males. This study demonstrated substantial increases in shoulder arthroplasties in Japan over the past decade, with notable differences in age and sex distributions compared with conventional RCR.

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Rotator cuff tears (RCTs) are among the most common shoulder disorders, particularly affecting middle-aged and elderly patients, leading to pain and functional disability.[18,40] In Japan, the aging population has increased the incidence of RCT, making effective management a critical public health concern.[25,46] Conservative treatments are often the first line of management for RCT; however, surgical intervention becomes necessary when nonoperative management fails.[20,22,28] Although rotator cuff repair (RCR) is the most frequently performed surgical procedure for these injuries,[42] shoulder arthroplasty is often required in cases in which functional cuff repair is not feasible or in the presence of cuff tear arthropathy.[1,5,6]

Developed by Grammont et al in 1987, reverse total shoulder arthroplasty (rTSA) has revolutionized the treatment of complex

This study was exempt from Institutional Board approval and informed consent requirements due to its retrospective design and the use of legally anonymized public data.

*Corresponding author: Masataka Minami, MD, PhD, Department of Orthopedic Surgery, Mayo Clinic, 200 First St. SW, Rochester, MN 55905, USA.
E-mail address: odriscollservice@mayo.edu (M. Minami).

https://doi.org/10.1016/j.jseint.2025.05.013
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shoulder conditions,[9] demonstrating excellent clinical outcomes across various patient populations, especially those with massive irreparable RCT and with cuff tear arthropathy.[6] Owing to its efficacy, the adoption of rTSA has increased dramatically worldwide, particularly in the United States.[2-4,8] In Japan, rTSA was approved much later in 2014, which may have influenced the surgical trends for RCT surgeries.[38]

According to an annual report by the Japanese Society for Replacement Arthroplasty (JSRA), the number of rTSA in Japan has increased from 630 in 2014 to 4,679 in 2022, accounting for over 84% of all shoulder arthroplasties in recent years.[10,12] Despite this remarkable increase, there is limited data on how the introduction of rTSA has impacted the overall trends in RCRs and shoulder arthroplasties in Japan.

The Ministry of Health, Labour, and Welfare in Japan operates the NDB Open Data Japan, one of the most expansive health-care databases on a national level globally, encompassing more than 95% of all insurance claims.[14,16,17,26]

This study aimed to analyze the annual trends in shoulder arthroplasties and RCRs from 2014 to 2022 using data from the national database to determine any significant changes in trends for rotator cuff surgery or shoulder arthroplasty in Japan. We also examined differences by sex and age group to provide a comprehensive overview of the evolving landscape of RCT surgeries in Japan. We hypothesized that the advent of rTSA has substantially altered treatment options for RCTs, affecting not only the frequency of arthroplasty procedures but also RCR.

Methods

Study design

We conducted a retrospective study using the following 2 databases. We mainly used the NDB Open Data Japan, public data with compiled claims information, operated by the Ministry of Health, Labour, and Welfare in Japan.[26] The NDB guidelines stipulate that fewer than 10 procedures per year should be masked to ensure the confidentiality of the patients and avoid the identification of personal information. To supplement more detailed data, we also used the annual report of the JSRA.[12]

Data collection

Data were retrieved from the NDB Open Data Japan website for 2014-2022. We used Microsoft Excel files titled "Number of calculations by division, sex, and age group" under "Surgery (code K). The numbers of outpatient and inpatient surgeries were summed for each year. We included 1) shoulder arthroplasty and 2) RCR as surgeries for RCT. Humerus head replacement (hemi shoulder arthroplasty), removal of arthroplasty, revision of arthroplasty, and arthroplasty for joints other than the shoulder were excluded. Finally, we extracted data for the following surgical procedures (Table I):

1) Total shoulder arthroplasty
   - Arthroplasty (code K082)
2) RCR
   - RCR (code K080-3),
   - Arthroscopic RCR (code K080-4).

Both categories of RCR surgery include simple and complex procedures. In general, RCR includes cuff tear repair regardless of additional procedures, such as tenotomy and tenodesis of the long head of the biceps but does not include only débridement, synovectomy, acromioplasty, and bursectomy. Complex RCRs are defined as procedures that are performed for cuff tears of 5 cm or more, involving fascia transplantation or muscle transfer, superior capsular reconstruction, patch replacement, and muscle advancement. In NDB Open Data Japan, shoulder arthroplasty combines both anatomic total shoulder arthroplasty (aTSA) and rTSA. To distinguish between rTSA and aTSA, we extracted data from the annual report of the replacement arthroplasty national registry published on the JSRA website.[12] JSRA data were derived from reports performed directly by orthopedic surgeons and could have distinguished between rTSA and aTSA only since 2016.

Table I
Associated surgical codes used in this study.

Surgical code	Classification	Specification
K080-3	Rotator cuff repair	Simple Complex
K080-4	Arthroscopic rotator cuff repair	Simple Complex
K082	Arthroplasty	Arthroplasty (shoulder)

Rotator cuff repair includes cuff tear repair regardless of additional procedures, such as tenotomy and tenodesis of the long head of the biceps, but does not include only débridement, synovectomy, acromioplasty, and bursectomy.
Complex rotator cuff repair is defined as the procedure that is performed for cuff tears of 5 cm or more, involving fascia transplantation or muscle transfer, superior capsular reconstruction, patch replacement, and muscle advancement.

Data analysis

For each surgical procedure, shoulder arthroplasty and RCR, we extracted the following:

- Total annual numbers: The total number of surgeries performed yearly from 2014 to 2022.
- Annual numbers by sex: The number of surgeries performed annually stratified by sex (male and female).
- Annual numbers by age group: The number of surgeries performed annually, categorized into 5-year age intervals (eg, 50-54 and 55-59 years).

Age-specific incidence rates were calculated by dividing the number of surgeries in each age group by the 2015 Japan Standard Population, based on the Ministry of Internal Affairs and Communications website data.[33,39]

Statistical analysis

Descriptive statistics were used to summarize the data and identify the annual trends in surgical procedures. The number of surgeries between the surgical types (shoulder arthroplasty vs RCR, open RCR vs arthroscopic RCR, and simple RCR vs complex RCR) were compared using independent *t*-tests. The mean age for each surgery type was calculated based on the age group distributions and compared between sexes and surgery types using an independent *t*-test. Age-specific incidence rates were expressed as the number of surgeries per 100,000 people. Linear regression analysis assessed trends in the number of surgeries and mean age within the groups over time. We used data from 2014 to 2022 to project the shoulder arthroplasty and RCR counts and incidence rates through 2030. Linear regression models (Python 3.13 and scikit-learn 1.5.2; Python Software Foundation, Wilmington, DE, USA) projected gradual increases for the younger age groups in both procedures.[31] For shoulder arthroplasty, conservative adjustments were applied to older age groups (75+ years) to control growth due to saturation trends, while standard linear models were used for RCR across all age groups. The remaining statistical analyses were conducted using the Statistical Package for the Social Sciences (version 29.0; IBM Corp., Armonk, NY, USA). For independent *t*-tests, the results were

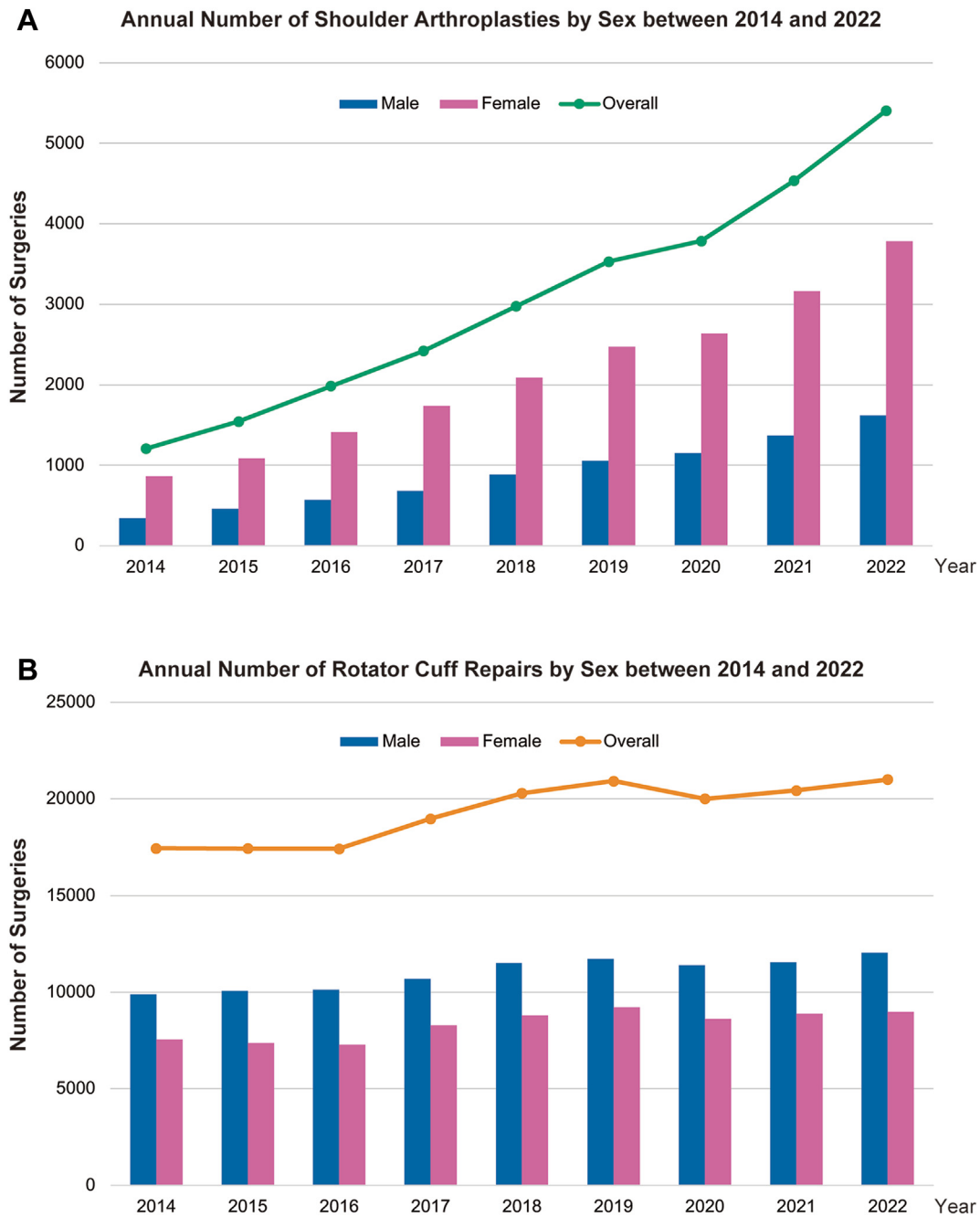

Figure 1 Annual number of surgeries by sex in Japan between 2014 and 2022. (**A**) The line and bars for each year indicate the number of shoulder arthroplasty surgeries. The *blue bars* indicate the number of surgeries for males, the *pink bars* indicate the number of surgeries for females, and the *green line* indicates the total number of surgeries for males and females combined. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (**B**) The line and bars indicate the number of rotator cuff repair surgeries. The *blue bars* indicate the number of surgeries for males, the *pink line* indicates the number of surgeries for females, and the *orange line* indicates the total number of surgeries for males and females combined. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

presented using mean differences and their 95% confidence intervals (CIs). For linear regression analyses, the results were reported using coefficients and their 95% CI.

Results

Annual number of surgeries

According to the NDB Open Data Japan, the number of shoulder arthroplasties increased dramatically each year from 1,246 in 2014 to 5,439 in 2022, more than fourfold (annual increase: 531 surgeries; 95% CIs: 530-532). (Fig. 1, *A*, Table II). Our projections indicate that this upward trend will continue, with procedures expected to reach approximately 7,644 by 2027 and 9,166 by 2030, driven primarily by rising demand in patients over 60 years of age (Fig. 2). According to the annual report of the JSRA, the percentage of arthroplasties that were rTSA was 94%, 87%, 85%, and 85% from 2016 to 2019, respectively (Fig. 3, *A*). From 2020 to 2022, the number of rTSA surgeries increased by nearly 800 surgeries each year to 3,074, 3,850, and 4,679, and the percentage of rTSA surgeries increased gradually to 84%, 86%, and 87%, respectively.

Table II
Annual number of shoulder arthroplasty and rotator cuff repair based on the NDB Open Data Japan between 2014 and 2022.

Procedure	2014	2015	2016	2017	2018	2019	2020	2021	2022	Total
Shoulder arthroplasty	1,246	1,578	2,009	2,454	3,008	3,548	3,834	4,559	5,439	27,675
Total rotator cuff repair	17,553	17,531	17,562	19,065	20,412	21,058	20,144	20,565	21,183	175,073
Open rotator cuff repair	2,923	2,836	2,528	2,503	2,393	2,188	2,053	1,932	1,825	21,181
Simple	2,349	2,364	2,091	2,101	2,028	1,855	1,740	1,657	1,584	17,769
Complex	574	472	437	402	365	333	313	275	241	3,412
Arthroscopic rotator cuff repair	14,630	14,695	15,034	16,562	18,019	18,870	18,091	18,633	19,358	153,892
Simple	12,490	12,918	13,269	14,671	15,986	16,732	16,159	16,693	17,710	136,628
Complex	2,140	1,777	1,765	1,891	2,033	2,138	1,932	1,940	1,648	17,264

NDB, the National Database of Health Insurance Claims and Specific Health Checkups.

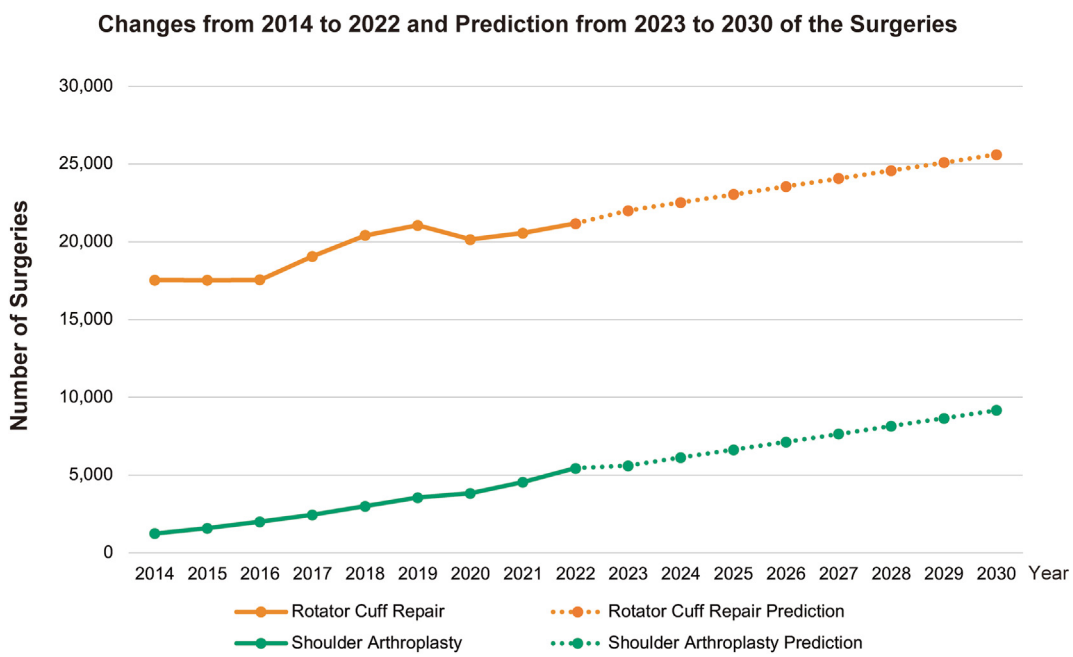

Figure 2 The number of rotator cuff repairs and shoulder arthroplasty from 2014 to 2022 and the prediction from 2023 to 2030. The *solid line* indicates the actual number of surgeries from 2014 to 2022, and the *dotted line* indicates the prediction for 2023 to 2030. The *orange line* indicates rotator cuff repair and the *green indicates* shoulder arthroplasty. The prediction was based on changes in the number of surgeries and Japanese population distribution data from 2014 to 2022. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

The overall number of RCRs did not decrease as rTSA became more popular, but actually increased gradually steadily from 17,553 in 2014 to 21,183 in 2022 (Fig. 1, *B*). The total number of RCRs was significantly greater than the number of shoulder arthroplasties (mean difference: 16,378; 95% CI: 14,901-17,854). The overall breakdown of RCRs showed that arthroscopic surgeries were more common than open surgeries in all years, with at least 4 times as many surgeries performed each year (mean difference: 14,746; 95% CI: 13,276-16,216) (Fig. 3, *B*, Table II). The number of arthroscopic RCRs increased gradually from 14,630 in 2014 to 19,358 in 2022 (annual increase: 643 surgeries; 95% CI: 642-644), whereas the number of open RCRs declined from 2,923 in 2014 to 1,825 in 2022 (annual decrease: 140 surgeries; 95% CI: 140-141). Comparing simple and complex RCR, simple RCR has been gradually increasing, with an annual increase of 565 surgeries (95% CI: 386-745) (Fig. 4, *A*, Table III). On the other hand, the number of complex RCR surgeries gradually tended to decrease from 2,714 in 2014 to 1,889 in 2022, although this trend was not statistically significant (annual decrease: 52 surgeries; 95% CI: −4 to 109). Due to this opposite trend, the proportion of complex RCR surgeries in the total number of RCR surgeries significantly decreased from 16% to 9% (annual decrease: 0.6%; 95% CI: 0.4-0.8) (Fig. 4, *B*). The projections indicate that the total number of RCRs is increasing at a rate of 513 per year and will reach approximately 25,000 by 2030 (Fig. 2).

Surgeries stratified by sex

Over the 9-year period, the total number of shoulder arthroplasties performed was 8,139 (30%) in males and 19,253 (70%) in females. Shoulder arthroplasty was almost twice as common in females in all 9 years (mean difference: 1,235; 95% CI: 456-2,014) (Fig. 1, *A*).

However, during the 9-year period, the total number of RCRs was 98,992 (57%) in males and 75,018 (43%) in females. Throughout the 9-year period, RCRs were significantly more common in males (mean difference: 2,664; 95% CI: 1,887-3,440) (Fig. 1, *B*).

Surgeries stratified by age and sex

For shoulder arthroplasty, the most common age group for males and females over 9 years was 75-79, with a mean age of

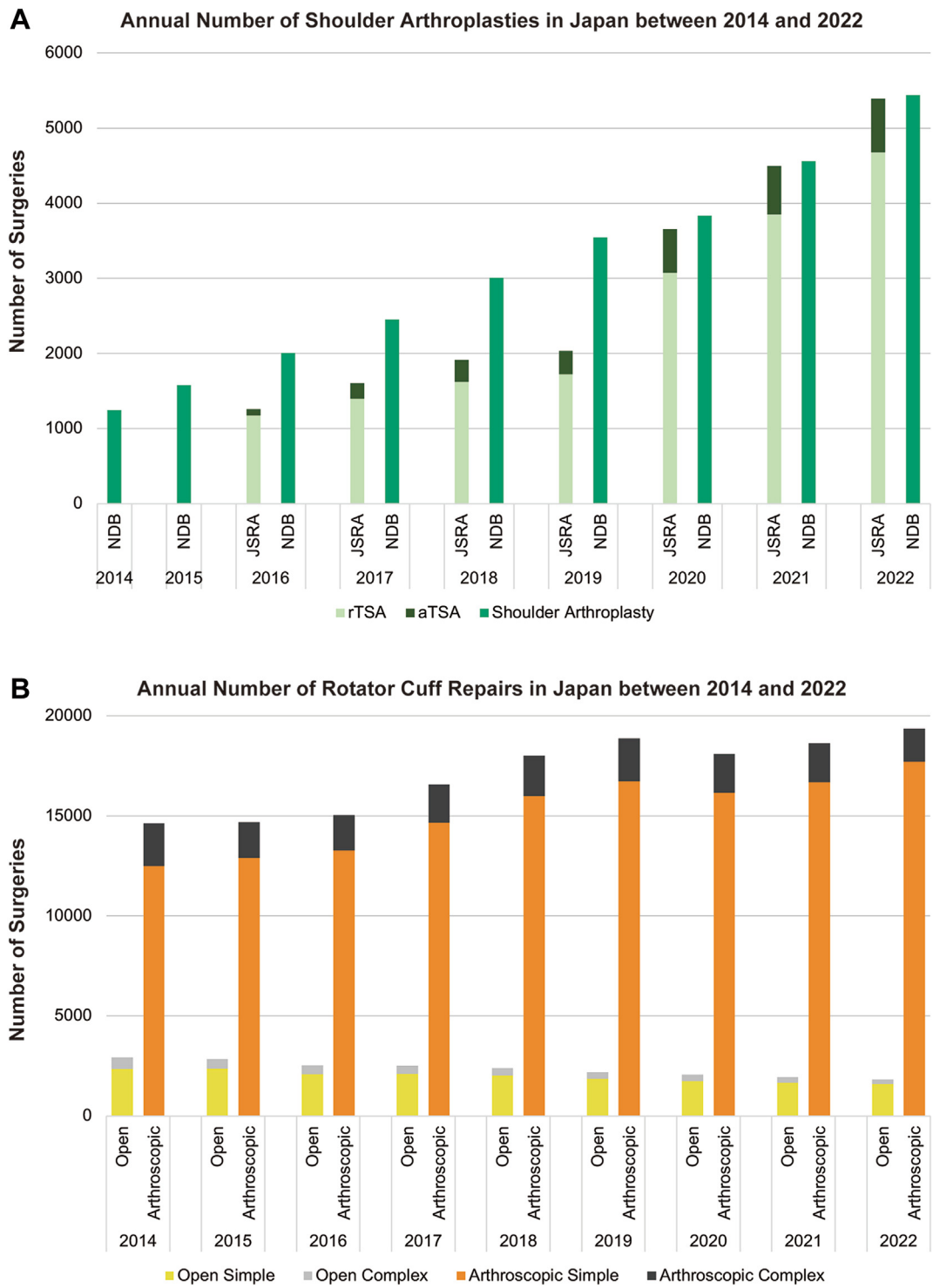

Figure 3 Annual number of surgeries in Japan between 2014 and 2022. (**A**) The bars for each year indicate the number of shoulder arthroplasty based on the 2 different databases. The *light green* and *dark green* bars on the *Left* indicate the number of reverse total shoulder arthroplasty (rTSA) and anatomic total shoulder arthroplasty (aTSA) based on the annual report of the Japanese Society for Replacement Arthroplasty (JSRA), respectively. The *green bars* on the *Right* indicate the number of shoulder arthroplasty, including rTSA and aTSA, based on the National Database of Health Insurance Claims and Specific Health Checkups (NDB). Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (**B**) The bars for each year indicate the number of rotator cuff repair surgeries. The *yellow* and *gray bars* on the *Left* indicate the number of open simple and complex surgeries, respectively, and the *orange* and *black bars* on the *Right* indicate the number of arthroscopic simple and complex surgeries, respectively. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

78 years (Fig. 5, *A*). In males, the mean age was 76 years, while in females, it was 79 years, with a high trend (mean difference: 2.8 years; 95% CI: 2.4-3.2). Looking at the 9-year change in the combined figures for males and females, the mean age of shoulder arthroplasty slightly increased from 77 years in 2014 to 78 years in 2022 (annual increase: 0.12 years; 95% CI: 0.07-0.17) (Fig. 6, *A*). The

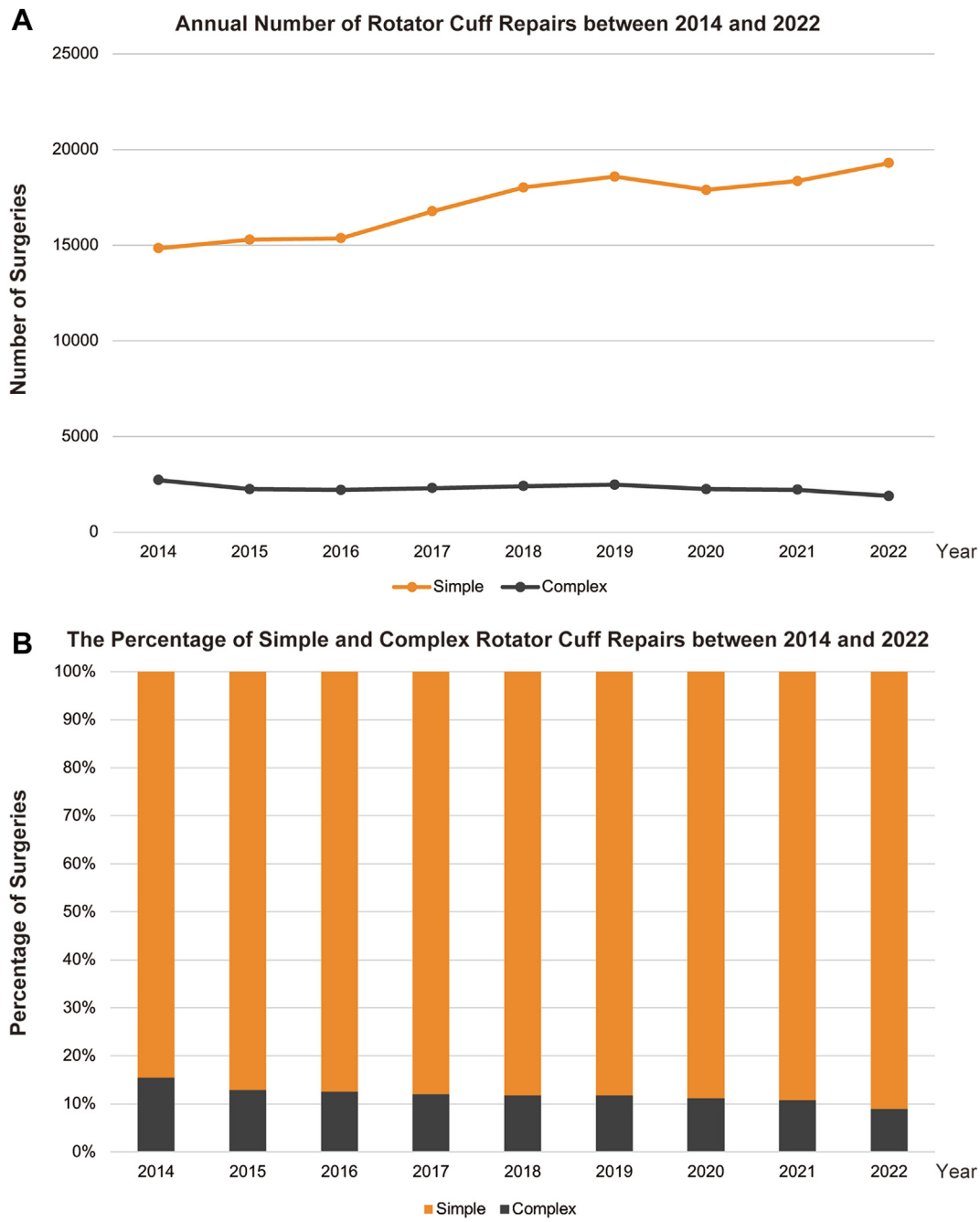

Figure 4 Annual number and percentage of rotator cuff repair in Japan between 2014 and 2022. (**A**) Annual number of rotator cuff repair in Japan between 2014 and 2022. The lines for each year indicate the number of rotator cuff repair surgeries. The *orange line* indicates the number of simple rotator cuff repair surgeries and the *black line* indicates the number of complex rotator cuff repair surgeries. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (**B**) The percentage of rotator cuff repair in Japan between 2014 and 2022. The bars for each year indicate the percentage of the 2 types of rotator cuff repair surgery in total rotator cuff repair surgery. The *orange bars* indicate the percentage of simple rotator cuff repair surgery, and the *black bars* indicate the percentage of complex rotator cuff repair surgery. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

9-year trend shows a uniform increase in the number of shoulder arthroplasties in all age groups.

The most common age group of RCR over the 9-year period was 70-74 years, with a mean age of 67 years (Fig. 5, *B*). In males, the most common age group of RCR over a 9-year period was 65-69 years, with a mean age of 65 years, while in females, the most common age group of RCR was 70-74 years, with a mean age of 68 years, showing a tendency for females to be older (mean difference: 3.3 years; 95% CI: 3.1-3.4). Looking at the 9-year change in the combined figures for males and females, the average age was 66 years in 2014 and 67 years in 2022, without significant change in the average age (annual increase: 0.02 years; 95% CI: −0.01 to 0.05) (Fig. 6, *B*). There was an increasing trend in the 50-54, 55-59, 70-74, 75-79, and 80-84 age groups, while there was little change in the 60-64 age group, and the number of surgeries in the 65-69 age group has been declining since 2020.

Table III
The number of surgeries and incidence of simple and complex rotator cuff repair between 2014 and 2022.

Procedure and age group	2014		2015		2016		2017		2018		2019		2020		2021		2022	
	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.
Simple rotator cuff repair	14,839	11.8	15,360	12.2	15,360	12.3	16,772	13.4	18,014	14.4	18,587	14.8	17,899	14.3	18,350	14.6	19,294	15.4
Complex rotator cuff repair	2,714	2.2	2,249	1.8	2,202	1.8	2,293	1.8	2,398	1.9	2,471	2.0	2,245	1.8	2,215	1.8	1,889	1.5

No., number; *Inc.*, incidence.
Incidence rates represent the number of surgeries per 100,000 people and were calculated by dividing by 2015 Japan standard population.

Age-specific incidence

The incidence of surgery per 100,000 people in each age group is shown in Tables III and IV. Shoulder arthroplasty was performed most frequently in the 75-79 or 80-84 age group, with an incidence as high as 29 per 100,000 population by age group. RCR was performed most frequently in the 70-74 age group, with an incidence of 44-57 per 100,000 people throughout the 9 years. The incidence of RCRs in the total population remained between 14 and 17 for RCR (Fig. 7). In contrast, it was much lower for shoulder arthroplasty than for RCR (mean difference: 13.0; 95% CI: 11.7-14.2), although the trend increased from 1 to 4. Across all 9 years, males had a higher incidence of RCR (mean difference: 5.0; 95% CI: 3.7-6.4), while females had a higher incidence of shoulder arthroplasty (mean difference: 1.8; 95% CI: 0.6-3.0).

Discussion

This nationwide analysis of shoulder surgery trends in Japan from 2014 to 2022 reveals 2 main findings. First, shoulder arthroplasty procedures demonstrated a substantial rise, more than quadrupling throughout the period. This was predominantly due to the increase in utilization of rTSA. Secondly, this increase in rTSA was not reflected by a decrease in RCR surgeries. The annual number of RCR surgeries, including primary repairs and joint-preserving procedures, steadily increased during the same time frame, while the proportion of complex RCR procedures showed a decreasing trend over this period.

Taken together, these findings suggest 2 things. The first is that the indications for rTSA expanded after its introduction in Japan to conditions beyond cuff tear arthropathy and irreparable massive cuff tears. This has been recognized elsewhere in the world, where rTSA is becoming preferred over aTSA for revision arthroplasty,[34] sequelae of proximal humeral fractures,[29] and for Walch B2 glenoid defects.[32]

The second implication is that the steady increase in RCR, particularly simple RCR, aligns with global trends and probably reflects advancements in minimally invasive surgical techniques, improved patient outcomes, and increased surgeon proficiency and training.[13,15,44] The fact that the proportion of complex RCRs observed in this study tended to decrease over the 9-year period is consistent with the increased utilization of rTSA.

Several factors may explain the observed trends. Japan presents a unique scenario due to the delayed approval of rTSA until 2014, in contrast to the United States and Europe, where rTSA has been widely adopted since the early 2000s. Katano et al analyzed the trends in arthroplasty in Japan between 2014 and 2017, including joints other than the shoulder.[14] In contrast to other joint replacement procedures, the annual number of shoulder arthroplasties increased significantly. The authors suggest that the arrival of rTSA may have influenced the dramatic increase in shoulder arthroplasties observed in this study.

The sustained increase in rTSA likely stems from its proven efficacy in treating cuff tear arthropathy and irreparable RCT, which are more prevalent in the aging population. Japan's aging demographic, with a growing elderly population, may contribute to the increased demand for arthroplasty.[23] Age-specific trends in our study revealed that RCRs were most frequent in the 70-74 age group, while arthroplasties were most common in patients aged 75-79 years. The findings demonstrated that rTSA is becoming a preferred option for older patients, likely due to its efficacy in treating cuff tear arthropathy and other complex conditions characteristic of these age groups.

Another important finding of this study is related to sex differences in surgical procedure distribution. We found that RCR was consistently performed more frequently in males, accounting for 57% of all procedures. This finding was consistent with the study by Li et al, who also found that 57% of RCR was performed in males.[19] This result could be explained by the higher incidence of RCT found in males in previous studies.[11,46] But it is also possible that RCRs in postmenopausal women are considered more likely to fail, and so are not as likely to be recommended for repair.[7,48]

In contrast, the distribution of patients undergoing shoulder arthroplasty procedures was higher in females (70%) than in males (30%). In addition, patients treated with shoulder arthroplasties (78 years) were generally older than those undergoing RCR (67 years). These results are consistent with that of Petel et al, who found that 59% of patients undergoing total hip or knee arthroplasty in the United States were female and significantly older than males.[30] This may be related to the fact that females had longer life expectancy and the female population is more significant among older people.[37,41] It might also be related to patients or surgeons having a higher threshold for joint replacement in men, especially if they are still quite active.

Our findings reveal a markedly lower incidence of RCR and shoulder arthroplasty in Japan compared to other countries. In the United States, RCR incidence per 100,000 people was 165 between 2007 and 2016 and is increasing.[47] In Europe, Italy reported 62 per 100,000 over a 14-year period,[36] and Latin America showed an increase from 25 to 49 per 100,000 from 2008 to 2018.[43] Shoulder arthroplasty incidence was 197 per 100,000 in the United States in 2017,[8] while multiple European countries averaged 20 per 100,000 in 2014.[21] In Japan, RCR and shoulder arthroplasty incidence rates remain much lower (15 and 2 per 100,000 in 2017, respectively). This may be due to different approaches in medical decision-making, rehabilitation systems, and insurance structures.[35,45] In addition, Japanese surgeons predominantly utilized joint-preserving surgeries for complex shoulder conditions until rTSA became available.[38] In such an environment, distinctive surgeries such as patch autograft,[27] superior capsule reconstruction,[24] and muscle advancement[28]

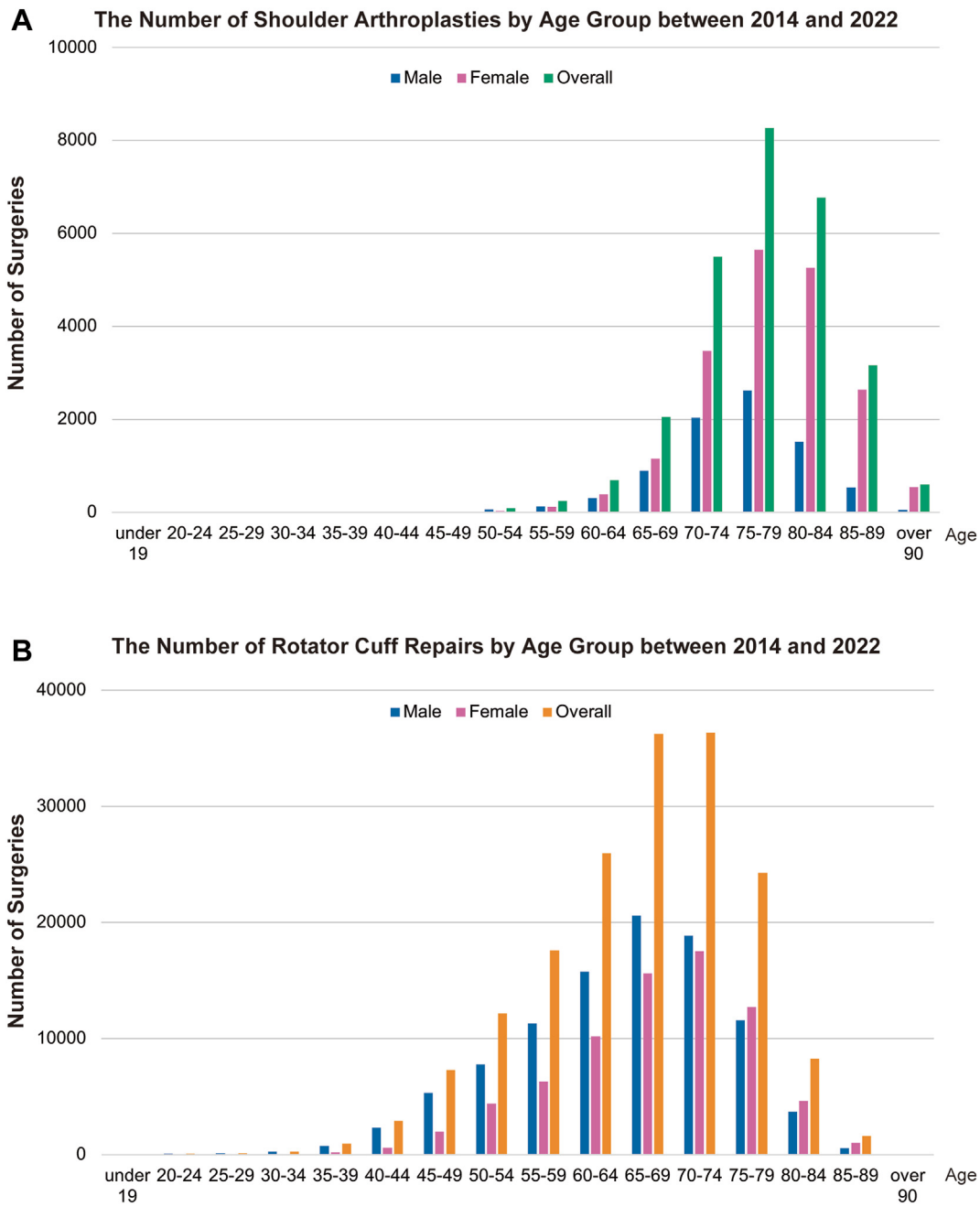

Figure 5 Total numbers of surgeries by age groups between 2014 and 2022. (**A**) The bars for each age group indicate the number of shoulder arthroplasty surgeries. *Blue bars* indicate the number of surgeries for males, *pink bars* indicate the number of surgeries for females, and *green bars* indicate the total number of surgeries for males and females combined. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (**B**) The bars for each age group indicate the number of rotator cuff repair surgeries. *Blue bars* indicate the number of surgeries for males, *pink bars* indicate the number of surgeries for females, and *orange bars* indicate the total number of surgeries for males and females combined. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

included in complex RCR have been developed. Our study predicts a future rise in shoulder arthroplasty surgeries in Japan, warranting close monitoring of potential shifts in complex RCR frequency.

Our findings suggest that the introduction of rTSA in Japan has broadened the surgical options for managing severe shoulder conditions, such as RCTs and shoulder arthropathy, especially in elderly patients. Clinicians can use these data to better inform patients of their surgical options, particularly when conservative or joint-preserving treatments fail. The remarkable growth in rTSA procedures, supported by encouraging clinical results in a diverse patient demographic and various conditions, is becoming a reliable alternative for appropriate patients.

This study has several limitations. First, this study relies on publicly available national data, which does not include detailed patient-specific information such as comorbidities, surgical outcomes, or postoperative recovery. Also, fewer than

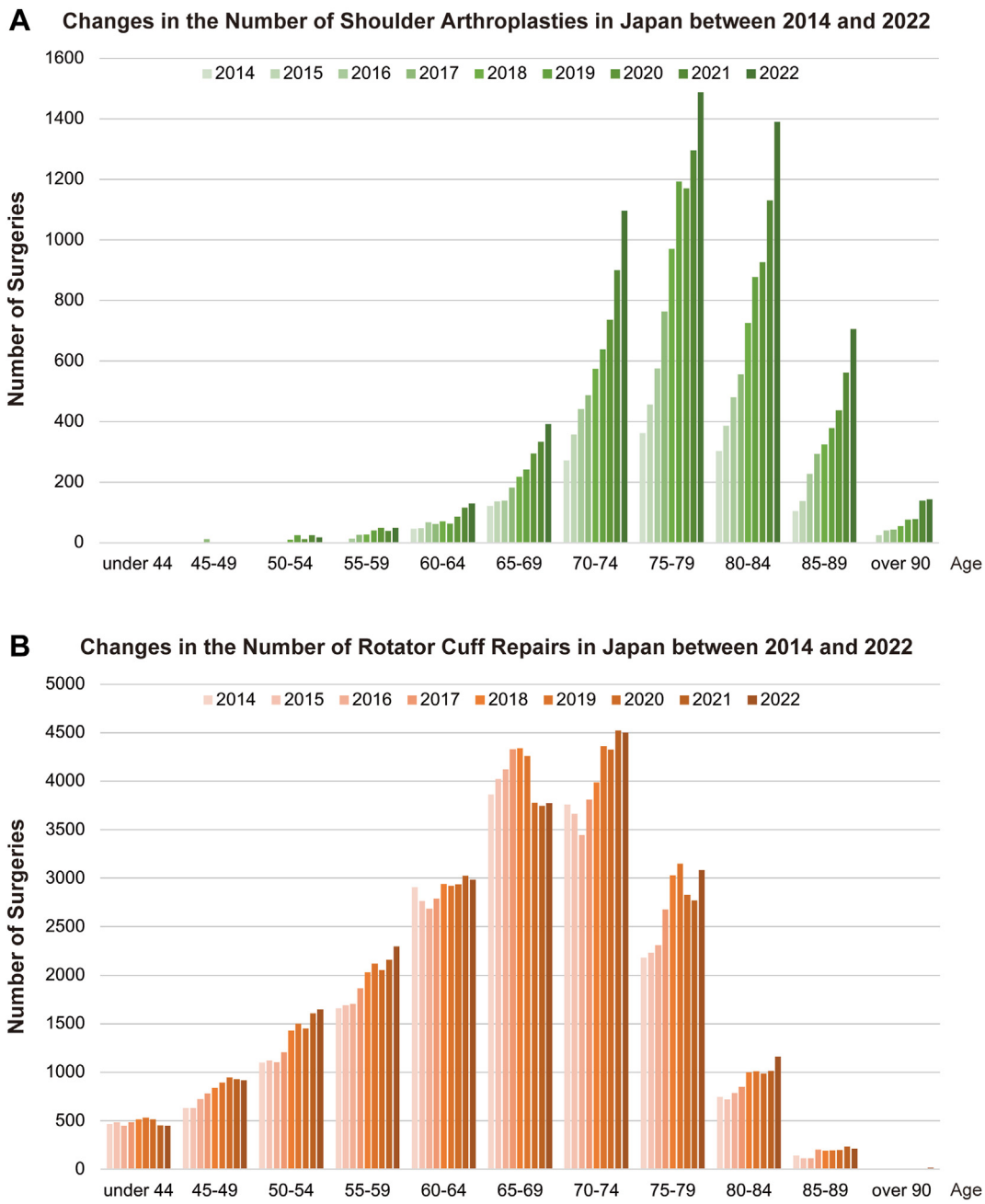

Figure 6 Changes in the number of surgeries in Japan between 2014 and 2022. (**A**) The *green bars* indicate the number of shoulder arthroplasty. The number of surgeries in each age group from left to right from 2014 to 2022. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (**B**) The *orange bars* indicate the number of rotator cuff repairs. The number of surgeries in each age group from left to right from 2014 to 2022. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

10 surgical procedures are masked to secure patients' anonymity so that the numbers could be more accurate. In terms of age-specific incidence, the result of this study may be influenced by the aging trend in Japan. The higher life expectancy of females may also be relevant. In addition, the NDB Open Data Japan include both rTSA and aTSA in shoulder arthroplasty. Although the annual report of the JSRA can distinguish between rTSA and aTSA, there was a significant discrepancy between the JSRA report and the actual NDB open data from 2016 to 2019. It is thought that the JSRA report during this period did not entirely reflect the exact number reported by orthopedic surgeons. Finally, although the increase in rTSA coincides with a decrease in complex RCR surgeries, suggesting a correlation, we could not calculate the specific percentage of rTSA performed due to RCT-derived conditions. As a result, the observed trend may overestimate the actual impact of rTSA on the number of complex RCR procedures. However, since the JSRA annual report states that 58%-69% of diagnoses of rTSA surgeries were due to irreparable RCT or cuff tear arthropathy, it is reasonable to assume that more than half of shoulder arthroplasty

Table IV
The number of surgeries and age-specific incidence of shoulder arthroplasty and rotator cuff repair between 2014 and 2022.

Procedure and age group	2014		2015		2016		2017		2018		2019		2020		2021		2022	
	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.	No.	Inc.
Shoulder arthroplasty																		
Under 44	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
45-49	0	0.0	0	0.0	0	0.0	11	0.1	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
50-54	0	0.0	0	0.0	0	0.0	0	0.0	10	0.1	24	0.3	12	0.1	24	0.3	17	0.2
55-59	0	0.0	0	0.0	13	0.1	26	0.3	28	0.3	40	0.5	49	0.6	39	0.4	49	0.6
60-64	46	0.5	48	0.5	68	0.7	62	0.7	70	0.8	63	0.7	86	0.9	116	1.3	129	1.4
65-69	121	1.3	136	1.5	139	1.5	182	2.0	217	2.3	241	2.6	295	3.2	333	3.6	391	4.2
70-74	272	3.4	357	4.5	442	5.6	487	6.2	574	7.3	638	8.1	735	9.3	900	11.4	1,096	13.9
75-79	361	5.7	456	7.2	576	9.1	762	12.1	970	15.4	1,192	18.9	1,170	18.6	1,295	20.5	1,486	23.6
80-84	303	6.4	386	8.2	480	10.2	555	11.8	726	15.4	877	18.6	926	19.6	1,129	23.9	1,389	29.4
85-89	104	3.3	137	4.4	227	7.2	293	9.3	325	10.4	379	12.1	437	13.9	561	17.9	705	22.5
Over 90	0	0.0	25	1.3	40	2.0	43	2.2	55	2.8	76	3.9	78	4.0	139	7.1	143	7.3
Total	1,246	1.0	1,578	1.3	2,009	1.6	2,454	2.0	3,008	2.4	3,548	2.8	3,834	3.1	4,559	3.6	5,439	4.3
Rotator cuff repair																		
Under 44	465	0.8	484	0.8	446	0.8	483	0.8	514	0.9	532	0.9	513	0.9	451	0.8	445	0.8
45-49	628	7.7	630	7.8	722	8.9	780	9.6	839	10.3	894	11.0	948	11.7	927	11.4	913	11.3
50-54	1,100	13.0	1,119	13.2	1,101	13.0	1,206	14.3	1,428	16.9	1,499	17.7	1,449	17.1	1,606	19.0	1,648	19.5
55-59	1,661	18.9	1,692	19.2	1,706	19.4	1,865	21.2	2,029	23.1	2,117	24.1	2,053	23.3	2,157	24.5	2,297	26.1
60-64	2,909	31.8	2,765	30.3	2,684	29.4	2,789	30.5	2,941	32.2	2,923	32.0	2,933	32.1	3,024	33.1	2,985	32.7
65-69	3,864	41.8	4,021	43.5	4,121	44.6	4,328	46.8	4,341	47.0	4,259	46.1	3,778	40.9	3,744	40.5	3,771	40.8
70-74	3,758	47.6	3,663	46.4	3,442	43.6	3,810	48.3	3,987	50.5	4,362	55.3	4,325	54.8	4,523	57.3	4,501	57.0
75-79	2,183	34.6	2,230	35.4	2,311	36.6	2,674	42.4	3,030	48.0	3,149	49.9	2,827	44.8	2,769	43.9	3,082	48.9
80-84	747	15.8	717	15.2	786	16.7	848	18.0	1,001	21.2	1,010	21.4	988	20.9	1,011	21.4	1,160	24.6
85-89	139	4.4	114	3.6	112	3.6	204	6.5	189	6.0	194	6.2	196	6.3	232	7.4	209	6.7
Over 90	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	13	0.7
Total	17,553	14.0	17,531	14.0	17,562	14.0	19,065	15.2	20,412	16.3	21,058	16.8	20,144	16.1	20,565	16.4	21,183	16.9

No., number; *Inc.*, incidence.
Incidence rates represent the number of surgeries per 100,000 people and were calculated by dividing by 2015 Japan standard population.

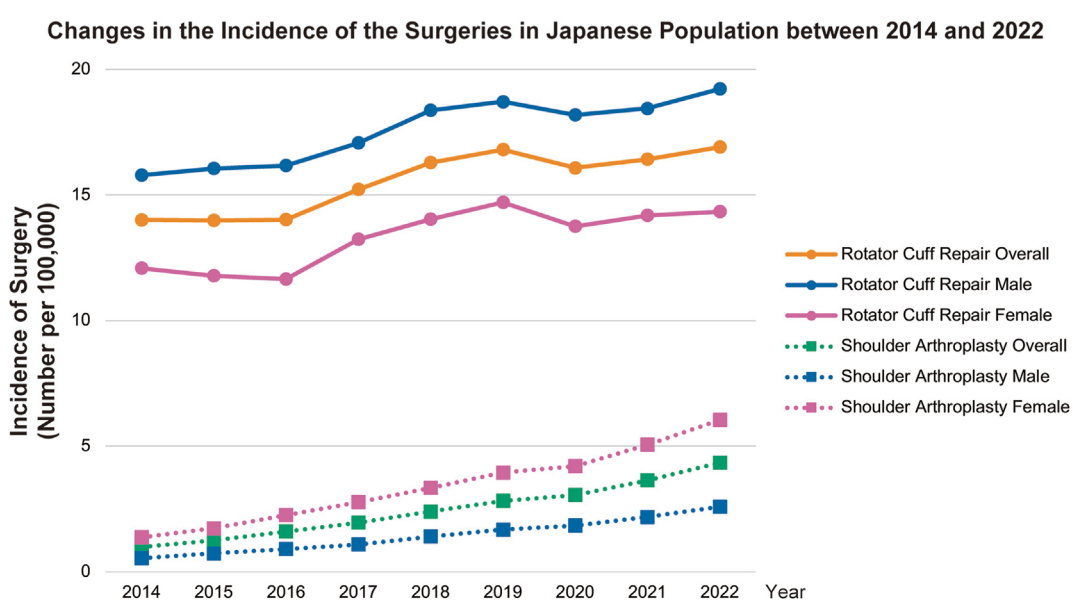

Figure 7 Changes in the incidence of the surgery in the Japanese population between 2014 and 2022. The *solid line* and *circle graphs* indicate rotator cuff repair, and the *dotted line* and *square graphs* indicate shoulder arthroplasty. *Orange* indicates rotator cuff repair, *green* indicates shoulder arthroplasty for males and females combined, and *pink* and *blue* indicate males and females for each procedure, respectively. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.

surgeries were consistently RCT-derived.[12] Future studies should incorporate more specific data to assess patient characteristics, diagnosis, and clinical outcomes in greater detail.

Conclusion

This study showed that shoulder arthroplasty increased dramatically over the 9 years from 2014 to 2022 following the introduction of rTSA in Japan, while RCR increased steadily but at a much lower rate. This study also revealed that shoulder arthroplasty was more common among females, RCR was more common among males, and the mean age at surgery was higher for shoulder arthroplasty than for RCR. These findings highlight the importance of shoulder arthroplasty, including rTSA, in expanding treatment options for complex shoulder disorders, particularly among older patients, while affirming the continued relevance of RCR.

Disclaimers:

Funding: No funding was disclosed by the authors.
Conflicts of interest: Shawn W. O'Driscoll, PhD, MD, and the research foundation with which he is affiliated received royalties from Acumed, LLC; otherwise, he and his immediate family did not receive any financial payments or other benefits from any commercial entity related to the subject of this article. All the other authors, their immediate families, and any research foundation with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

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